Training-Free Token Reduction on MotionBench

Results

HuVLLM Project

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Eleven training-free video-LLM token-reduction methods, evaluated on MotionBench (8,052 questions, 4,018 scoreable) across two backbones — LLaVA-OV-7B and Qwen3-VL-8B — at 32 frames. Results only: full methodology, the verification gate, and limitations are in `paper.tex`. At $n = 4{,}018$ the binomial 95% confidence half-width is ± 1.54 points; * marks a difference significant by McNemar's test on paired predictions ($\chi^2 \geq 3.84$, $p < 0.05$).

Contents

1 Common settings

Held fixed across every method and both backbones:

- **32 frames**, sampled uniformly per video.
- **Greedy decoding** (`do_sample=False`, `max_new_tokens=16`), batch size 1.
- **Letter-match scoring** (A–D) against ground truth; the 4,034 `NA` items are excluded, leaving 4,018 scoreable.
- Methods exposing a single retention knob are standardized to **15% nominal retention**; methods with no such knob (frame-selection methods, or methods whose budget is fixed internally) keep their published defaults — forcing 15% on those would break paper fidelity.

Per-method settings, mechanisms and measured token budgets are in Table 4 (§5).

2 Master table

Table 1 is every accuracy figure in this report in one place: both backbones, all six subcategories, both baselines, and every retention level actually run. Four of eleven methods (FastV, FlashVID, HoliTom, PruneVID) expose a comparable retention knob. FlashVID and HoliTom were run at 10/15/20/25% on LLaVA-OV (20% and every other point are the authors' own exact published settings) and 10/15/25% on Qwen3-VL; FastV was run at 10/15/25/50/75% on LLaVA-OV (the published range) and 10/15/25% on Qwen3-VL; PruneVID's authors publish exactly one operating point, `cluster_ratio=0.5` — its "retention" column is therefore 10/25/50%, not 10/15/25%, and its 50% cell, not any 15% cell, is what appears in the standardized-setting sections below (†). The remaining seven methods appear once, at their standardized or default setting (§1). § AIM's released code ships two active merge steps, so its LLaVA-OV cell runs at 25% retention, not 15%. Rounding is half away from zero throughout (see note, Table 1 caption).

Table 1: Master results table. AO = Action Order, CM = Camera Motion, LM = Location-related Motion, MR = Motion Recognition, MO = Motion-related Objects, RC = Repetition Count (% correct within category, chance = 25%). Δ is vs. that row's backbone baseline. * = significant. [†] PruneVID's only published operating point is `cluster_ratio=0.5`; its "standardized-setting" row is that 50% cell, not a 15% run — it was never run at 15% on either backbone. [‡] These two FlashVID/Qwen3-VL cells predate the authors'-code fix (§5) and are the incomplete reimplementation, not the verified method — not directly comparable to the 59.36% at 15% above. By a striking coincidence the reimplementation's own 25% cell (59.36%, −3.16) numerically matches the verified 15% cell exactly while sharing only 5,725 of 8,052 predictions with it; this is confirmed independent, not a duplicated run. Six Repetition Count cells at the 15% nominal setting land on an exact x.25 rounding tie (e.g. 101/400); this table rounds half away from zero, matching `paper.tex` — naive recomputation with default rounding disagrees on these cells by 0.1 point.

Method	Backbone	Retention	Overall	Δ	AO	CM	LM	MR	MO	RC
Baseline	LLaVA-OV-7B	—	*52.66*	—	40.5	45.2	55.5	57.0	71.2	23.8
Baseline	Qwen3-VL-8B	—	*62.52*	—	46.1	63.1	65.0	67.3	79.0	33.8
LLaVA-OV-7B, standardized setting										
DyCoke	LLaVA-OV-7B	15%	53.36	+0.70	38.9	48.8	55.7	58.2	70.9	25.3
FlashVID	LLaVA-OV-7B	15%	53.31	+0.65	39.9	45.7	53.8	58.0	71.7	28.2
HoliTom	LLaVA-OV-7B	15%	53.14	+0.47	40.8	49.6	52.6	57.0	71.4	27.3
MDP3	LLaVA-OV-7B	15%	53.06	+0.40	40.5	49.6	53.5	56.8	71.6	26.5
VideoITG	LLaVA-OV-7B	15%	52.86	+0.20	40.1	47.0	53.7	57.6	70.1	26.8
AIM	LLaVA-OV-7B	25%[§]	52.86	+0.20	41.4	48.1	54.2	57.0	71.9	22.3
STTM	LLaVA-OV-7B	15%	51.72	–0.95	39.9	48.1	53.8	53.5	70.6	28.8
PruneVID-OV	LLaVA-OV-7B	50%[†]	38.20	–14.46*	32.0	34.8	35.5	38.4	52.9	27.3
FastV	LLaVA-OV-7B	15%	36.78	–15.88*	32.8	31.9	34.1	35.7	53.8	25.3
Qwen3-VL-8B, standardized setting										
PruneVID	Qwen3-VL-8B	50%[†]	62.17	–0.35	46.1	62.9	64.1	67.1	79.1	32.5
DyCoke	Qwen3-VL-8B	15%	61.85	–0.67	45.1	62.6	63.9	66.4	78.4	34.5
STTM	Qwen3-VL-8B	15%	61.60	–0.92*	45.9	61.6	64.8	66.0	77.1	34.8
HoliTom	Qwen3-VL-8B	15%	60.33	–2.19*	44.7	61.3	61.0	66.4	76.2	29.0
MDP3	Qwen3-VL-8B	15%	59.66	–2.86*	44.9	64.7	62.8	64.0	77.4	23.0
FlashVID	Qwen3-VL-8B	15%	59.36	–3.16*	43.5	57.4	62.6	65.6	77.2	23.5
FastV	Qwen3-VL-8B	15%	59.01	–3.51*	46.2	61.0	61.9	63.5	72.2	30.5
VisionZip (contextual)	Qwen3-VL-8B	15%	58.81	–3.71*	43.2	57.4	61.9	64.2	74.3	29.5
VideoITG	Qwen3-VL-8B	15%	56.35	–6.17*	45.3	53.5	59.3	60.2	75.4	22.2
AIM	Qwen3-VL-8B	15%	55.97	–6.55*	45.9	56.6	59.5	58.3	72.2	27.0

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Table 1 (continued)

Method	Backbone	Retention	Overall	Δ	AO	CM	LM	MR	MO	RC
LLaVA-OV-7B, retention sweep (15% and PruneVID's 50% already listed above)										
FastV	LLaVA-OV-7B	10%	36.06	–16.60*	32.9	31.9	33.9	34.1	53.3	24.5
FastV	LLaVA-OV-7B	25%	36.73	–15.93*	32.0	31.7	33.3	36.3	53.0	26.0
FastV	LLaVA-OV-7B	50%	36.34	–16.33*	30.8	30.6	33.9	36.2	52.8	24.5
FastV	LLaVA-OV-7B	75%	35.69	–16.97*	30.6	30.1	32.2	35.9	52.6	22.3
FlashVID	LLaVA-OV-7B	10%	52.51	–0.15	41.0	46.8	53.3	56.3	70.7	26.5
FlashVID	LLaVA-OV-7B	20%	53.71	+1.05*	41.6	46.8	55.5	57.2	73.3	26.8
FlashVID	LLaVA-OV-7B	25%	53.36	+0.70	42.2	47.5	55.1	57.5	71.9	23.8
HoliTom	LLaVA-OV-7B	10%	52.76	+0.10	40.5	48.8	53.3	55.8	71.2	29.0
HoliTom	LLaVA-OV-7B	20%	53.06	+0.40	40.7	48.3	53.8	57.6	71.7	23.8
HoliTom	LLaVA-OV-7B	25%	53.41	+0.75	40.7	48.3	54.8	57.7	72.3	24.5
PruneVID-OV	LLaVA-OV-7B	10%	38.10	–14.56*	33.5	33.5	36.8	38.2	52.3	25.2
PruneVID-OV	LLaVA-OV-7B	25%	38.38	–14.29*	32.2	33.2	36.6	38.8	53.5	26.2
Qwen3-VL-8B, retention sweep (15% and PruneVID's 50% already listed above)										
FastV	Qwen3-VL-8B	10%	56.92	–5.60*	44.5	60.5	59.0	61.4	69.9	28.0
FastV	Qwen3-VL-8B	25%	60.60	–1.92*	46.4	62.3	63.7	65.2	74.1	32.8
FlashVID[‡]	Qwen3-VL-8B	10%	54.73	–7.79*	44.5	54.5	56.0	58.4	70.9	25.0
FlashVID[‡]	Qwen3-VL-8B	25%	59.36	–3.16*	43.9	62.3	61.0	63.8	75.7	29.8
HoliTom	Qwen3-VL-8B	10%	60.05	–2.46*	44.9	60.8	59.7	65.7	77.4	28.7
HoliTom	Qwen3-VL-8B	25%	60.43	–2.09*	45.1	61.0	61.0	66.0	77.5	28.7
PruneVID	Qwen3-VL-8B	10%	60.23	–2.29*	46.6	60.8	61.0	65.2	77.1	28.7
PruneVID	Qwen3-VL-8B	25%	61.40	–1.12*	46.1	60.0	63.0	66.6	78.7	31.5

3 Results by backbone

3.1 LLaVA-OV-7B

Table 2: LLaVA-OV-7B, 32 frames, 15% nominal retention except where noted. All nine methods that run on this backbone. § AIM runs at 25% retention (two active merge steps ship in the released code). † PruneVID-OV runs at `cluster_ratio=0.5` (50% retention) — the authors' only published operating point, never 15%.

Method	Overall	Δ	AO	CM	LM	MR	MO	RC
DyCoke	53.36	+0.70	38.9	48.8	55.7	58.2	70.9	25.3
FlashVID	53.31	+0.65	39.9	45.7	53.8	58.0	71.7	28.2
HoliTom	53.14	+0.47	40.8	49.6	52.6	57.0	71.4	27.3
MDP3	53.06	+0.40	40.5	49.6	53.5	56.8	71.6	26.5
VideoITG	52.86	+0.20	40.1	47.0	53.7	57.6	70.1	26.8
AIM§	52.86	+0.20	41.4	48.1	54.2	57.0	71.9	22.3
Baseline	*52.66*	—	*40.5*	*45.2*	*55.5*	*57.0*	*71.2*	*23.8*
STTM	51.72	–0.95	39.9	48.1	53.8	53.5	70.6	28.8
PruneVID-OV*†	38.20	–14.46	32.0	34.8	35.5	38.4	52.9	27.3
FastV*	36.78	–15.88	32.8	31.9	34.1	35.7	53.8	25.3

3.2 Qwen3-VL-8B

Table 3: Qwen3-VL-8B, 32 frames, 15% nominal retention except where noted. All ten portable methods. VisionZip is the contextual half only — Qwen3-VL's vision tower has no CLS token, so the dominant half is not implementable; not the published method. FlashVID and STTM run the authors' own code at published settings (§5); neither carries the off-spec caveat from earlier revisions of this report. † PruneVID runs at `cluster_ratio=0.5` (50% retention), the authors' only published point, never 15%.

Method	Overall	Δ	AO	CM	LM	MR	MO	RC
Baseline	*62.52*	—	*46.1*	*63.1*	*65.0*	*67.3*	*79.0*	*33.8*
PruneVID†	62.17	–0.35	46.1	62.9	64.1	67.1	79.1	32.5
DyCoke	61.85	–0.67	45.1	62.6	63.9	66.4	78.4	34.5
STTM*	61.60	–0.92	45.9	61.6	64.8	66.0	77.1	34.8
HoliTom*	60.33	–2.19	44.7	61.3	61.0	66.4	76.2	29.0
MDP3*	59.66	–2.86	44.9	64.7	62.8	64.0	77.4	23.0
FlashVID*	59.36	–3.16	43.5	57.4	62.6	65.6	77.2	23.5
FastV*	59.01	–3.51	46.2	61.0	61.9	63.5	72.2	30.5
VisionZip (ctx.)*	58.81	–3.71	43.2	57.4	61.9	64.2	74.3	29.5
VideoITG*	56.35	–6.17	45.3	53.5	59.3	60.2	75.4	22.2
AIM*	55.97	–6.55	45.9	56.6	59.5	58.3	72.2	27.0

4 Figures

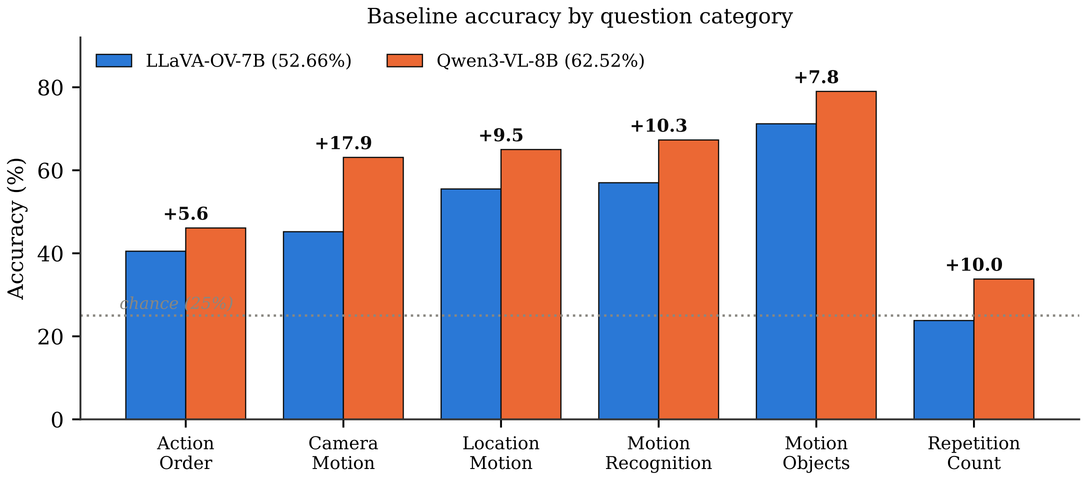

Figure 1: Baseline accuracy by category, both backbones (bar).

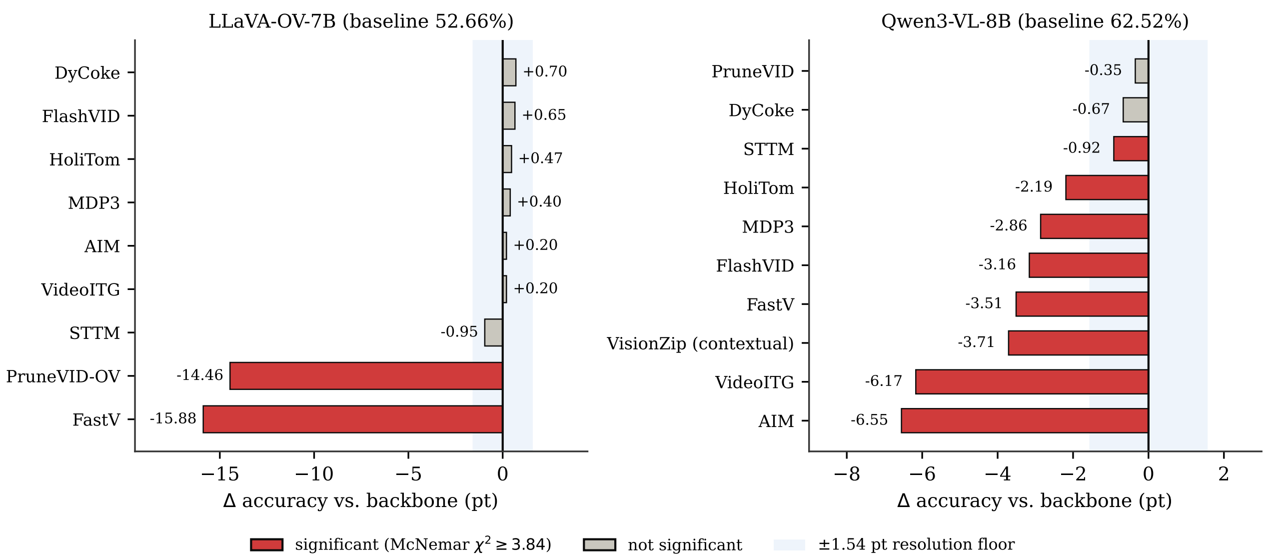

Figure 2: Change vs. backbone, every method, both backbones (bar). Red = significant by McNemar's test; the shaded band is the ±1.54-point resolution floor.

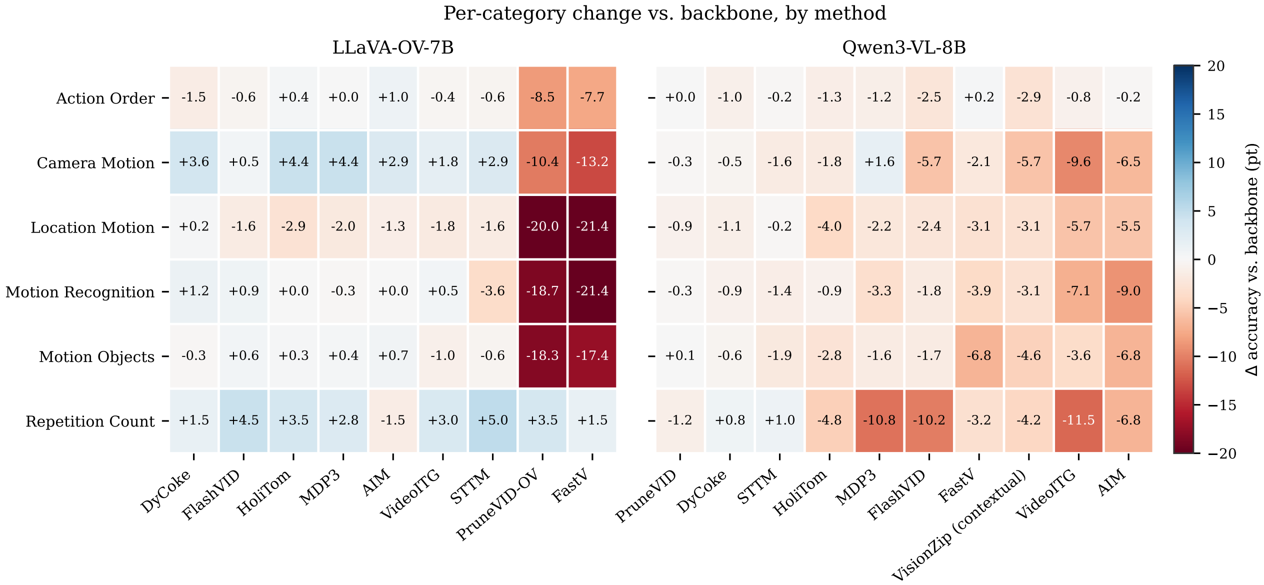

Figure 3: Per-category change vs. backbone, by method (heatmap). Blue = gain, red = loss.

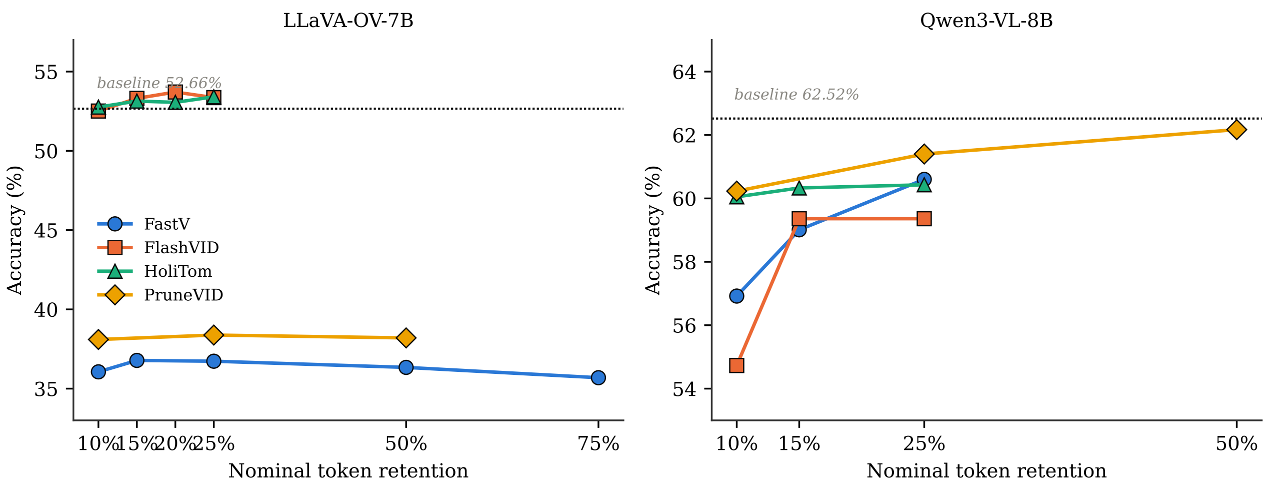

Figure 4: Accuracy against nominal token retention, the four methods with a retention knob (dot + line).

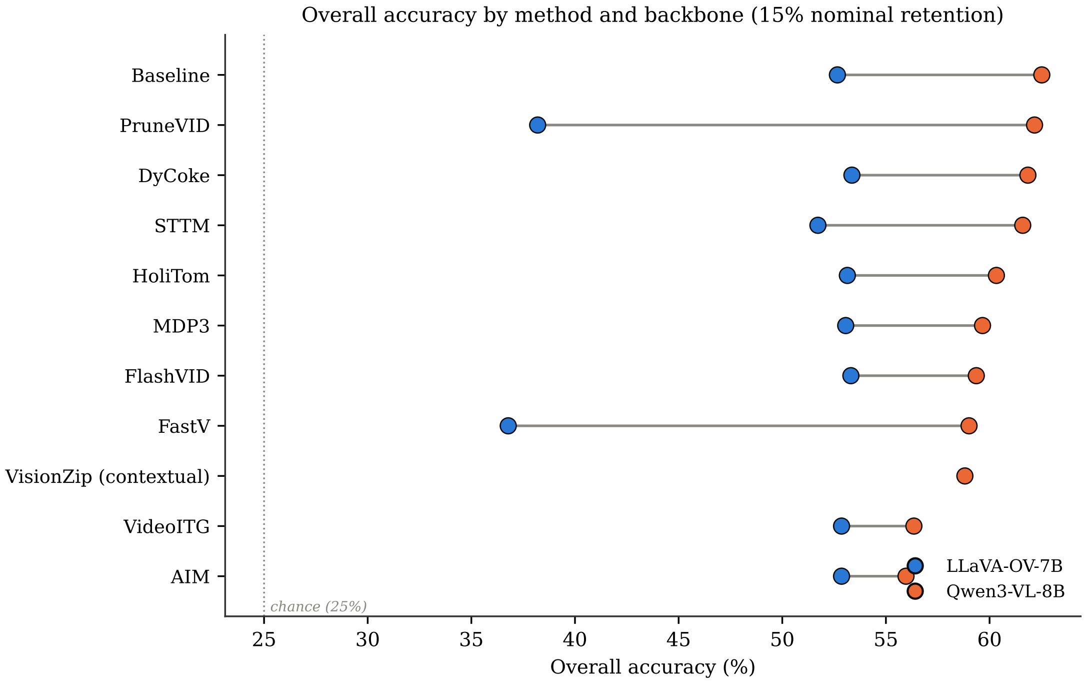

Figure 5: Overall accuracy by method and backbone at 15% nominal retention (Cleveland dot plot). Rows sorted by Qwen3-VL accuracy; the line spans each method's two backbone results.

5 Per-method settings

Table 4: Per-method mechanism, key parameters (as run in this report), and measured visual-token budget. Budgets are read from each method's execution log on Qwen3-VL-8B (11,664 visual input tokens at 32 frames), not from its CLI flag — nominal 15% describes the configured knob, not necessarily what is actually retained. Frame-selection methods have no token budget to measure; they instead reduce the number of frames fed to the model.

Method	Mechanism	Key parameters (this report)	Measured budget
HoliTom	pre-LLM segment retain + in-LLM merge	`RETAIN_RATIO=0.15 T=0.80 HOLITOM_k=18 HOLITOM_r=0.5`	7.5% (875 tok.)
STTM	quadtree spatio-temporal merge	LLaVA-OV: `thresh=0.85 temporal=0.65 root=1`. Qwen3-VL: `thresh=0.85 temporal=0.65 root=2` — a published setting (`s3_sttm_pub_run`), replacing an earlier off-spec `root=0, temporal=--1.0` cell	9.0% (1,050 tok.) on LLaVA-OV; not re-measured at the corrected Qwen3-VL setting
FastV	in-LLM attention rerank, layer 2	`K=2, R=0.85` (keep 15%) — outside the published K=2/R=50% range, though K=2/R=50% and K=2/R=75% were also run (§2) and land in the same collapse band	15.0% (1,750 tok.)
FlashVID	pre-LLM temporal-segment merge + inner-LLM compression, layer 28	authors' `flashvid()` package: `retention_ratio=0.15 alpha=0.7 temporal_threshold=0.8 token_selection_method=attn_div pruning_layer=28 llm_retention_ratio=0.1 expansion=1.25 min_segment_num=4`	15.0% (1,750 tok.) pre-LLM; the layer-28 stage compresses surviving visual tokens by a further 90% mid-network, not captured by this single figure
AIM	bipartite merge + PageRank prune	2-step merge schedule compiled in; no CLI retention knob. Qwen3-VL 15%; LLaVA-OV 25% (ships two active steps)	15.0% (1,750 tok.)
VisionZip (contextual)	key-similarity merge in the vision tower	`dominant=54 contextual=10` at 8 frames (native LLaVA-1.5-7B form); Qwen3-VL runs the contextual half only — no CLS token to select the dominant half	15.0% (1,750 tok.)
DyCoke	temporal merge + KV prune, layer 3	`l=3 p=0.7 k=0.7`; no single retention knob, net ∼49%	49.0% (5,716 tok.)
PruneVID	DPC-KNN cluster merge, layer 10	`cluster=0.50 seg=0.25`; LLaVA-OV port has no upstream reference (upstream targets PLLaVA only)	50.0% (5,832 tok.)
MDP3	frame selection (DPP)	`pool=32 select=8` — selects frames, not tokens	n/a (8 of 32 frames)
VideoITG	grounded frame selection	512 sampled → 32 selected	n/a (32 of 512 sampled)
DyTo†	FINCH clustering + ToMe token merge	`temporal_aggregation= spatial_tome_finch_dynamic_all_frms`; own Vicuna-7B backbone	∼3,680 tok. from 100 input frames

† DyTo does not run on either standardized backbone; listed here for completeness (see Table 6).

6 Correct/incorrect vs. baseline

Fixed = wrong under the baseline, correct under the method. *Broke* = correct under the baseline, wrong under the method. Both are McNemar discordant pairs on the 4,018 scoreable items. *Differ* counts predictions unequal to the baseline's over all 8,052 rows (0 would mean a silent no-op — see `paper.tex` for the verification gate this guards against).

Table 5: Paired-prediction statistics against each backbone's baseline.

Method	Differ	Fixed	Broke	χ^2	Significant
LLaVA-OV-7B					
DyCoke	1,031	170	142	2.34	no
FlashVID	1,610	268	242	1.23	no
HoliTom	1,838	299	280	0.56	no
MDP3	1,452	246	230	0.47	no
VideoITG	1,193	192	184	0.13	no
AIM	1,406	227	219	0.11	no
STTM	4,859	329	367	1.97	no
PruneVID-OV	4,536	473	1,054	220.30	yes
FastV	4,605	475	1,113	255.52	yes
Qwen3-VL-8B					
PruneVID	1,074	56	70	1.34	no
DyCoke	1,048	83	110	3.50	no
HoliTom	1,657	131	219	21.63	yes
MDP3	1,626	159	274	30.01	yes
FastV	2,029	149	290	44.65	yes
VisionZip (contextual)	2,035	191	340	41.25	yes
STTM	1,510	133	170	4.28	yes
FlashVID	2,069	157	284	36.00	yes
VideoITG	2,522	206	454	92.44	yes
AIM	2,861	194	457	105.44	yes
Backbone vs. backbone					
Qwen3-VL-8B vs. LLaVA-OV-7B	7,081	942	546	127.47	yes

Divergence volume does not predict damage. STTM on LLaVA-OV changes 4,859 predictions — more than FastV's 4,605 — yet loses 0.94 points and is not significant, because its changes fall almost evenly in both directions (329 fixed, 367 broken). FastV changes a comparable number destructively (475 fixed, 1,113 broken).

7 Methods on non-standard backbones

Table 6: Runs on backbones other than the two standardized ones. Not comparable to Tables 2–3; no Δ is given because no matched baseline exists for either row.

Method	Backbone	Overall	AO	CM	LM	MR	MO	RC
VisionZip	LLaVA-1.5-7B	40.09	33.7	33.0	37.2	41.1	57.4	25.8
DyTo (recon. TW-FINCH)	Vicuna-7B	42.06	32.4	33.0	42.1	43.0	61.7	26.0

8 Notable

The backbone swap (LLaVA-OV-7B $\rightarrow$ Qwen3-VL-8B) moves accuracy by 9.86 points — more than any token-reduction method moves it in either direction, on either backbone, at any retention level in Table 1. Method rankings do not transfer across backbones: PruneVID is the worst LLaVA-OV result (38.20%) and the best Qwen3-VL one (-0.35 vs. baseline); Figure 5 makes this visible directly — several lines cross. Two methods, FastV and PruneVID-OV, collapse specifically on LLaVA-OV (37–38%, more than 14 points below baseline and significant by McNemar's test), while every other LLaVA-OV method is statistically indistinguishable from the backbone. FastV's collapse was checked against its own published range: keep 50% and keep 75% (Table 1) land at 36.34% and 35.69%, the same collapse band as keep 15% — the failure is not an artifact of an unpublished setting. On Qwen3-VL the pattern inverts: every method loses accuracy and eight of ten losses are significant, yet none approaches the LLaVA-OV collapse in size — the largest Qwen3-VL loss (AIM, -6.55) is still smaller than the smallest LLaVA-OV collapse. Tripling the retention budget from 10% to 25% moves no LLaVA-OV method more than 0.85 points, but moves FastV 3.68 points on Qwen3-VL (Table 1, Figure 4). FlashVID's own equivalent Qwen3-VL span, 4.63 points, is not reported here because it is computed entirely from the pre-fix reimplementation (Table 1's ‡ note) and is not a claim about the verified method.

Three methods have an authors'-validated multi-point retention sweep — i.e., their own released code publishes more than one operating point, so moving the retention knob within that range is not extrapolation. **FlashVID** and **HoliTom** publish an identical four-point sweep, 10/15/20/25%, on every backbone they support; both were run at all four here (Table 1) and stay within 1.1 points of baseline at every setting on LLaVA-OV. **FastV** publishes 12.5/25/50/75% keep; 10–75% was run here, spanning and bracketing that range. The other five methods with any retention-like parameter do not qualify: **PruneVID** publishes exactly one ratio (0.5, Table 1's † note); **DyCoke** has no single retention knob, only one fixed three-parameter recipe; **AIM** has no knob at all on LLaVA-OV and an unpublished, self-ported one on Qwen3-VL; **STTM** and **VisionZip** are configured per-benchmark or as fixed token budgets, not a sweepable percentage.